FIGURE 2 CONTINUED · EXISTING MECHANISTIC EVIDENCE · SHEET 2/2

Mechanistic explanations and their corresponding measurements

Companion to manuscript Figures 2–3. Existing mechanistic evidence explains specific contributions and motivates measurements; the common gain/recovery panels do not causally partition all three cue-duration curves.

Model / candidate mechanism	Observed evidence	Corresponding measurements
Holmes 3 Coupled Rac/Rho domains and cue coverage permit a locally activated challenger to coexist with an incumbent. Broadening and domain competition remain important even when local gain and damage repair change little.	Fourfold inactive mobility fails to rescue fixed-support L28/L50 at 5120 units; relative-20% cues pass orientation at all eligible sizes. Ring placement separates merger from distinct sites within one complete C40 basal state; a C30 example has a cue-associated survivor.	Image both Rac-like and antagonistic activity regions; vary input position and support at fixed measured size. Track site extent, separation, merger and extinction through withdrawal alongside the global reporter vector and its magnitude.
Corrected deBelly Length-dependent resting mechanics and old-end MCA depletion delay incumbent suppression; at the longest lengths, slower local MCA accumulation adds a further delay. MCA-to-Rac inhibition connects this local organization to activity.	Relative-20% cues: L23→31 old-end MCA starts 0.05348→0.04823; by 180 s it reaches 0.06451→0.06044. The 10% Rac-decline clock is 123.49→142.97 s, versus 637.39 s at L39. Corrected resting tension-to-Rho ablation reduces displacement spread 50.4%.	Measure old/new-end activity, attachment-reporter level, local flow and protrusion together. Compare basal depletion and absolute common-time refill with incumbent decline; quantify local transport/replacement balance. Perturb tension coupling separately to test the starting-state mechanism.
Spring Extension-dependent global deactivation couples mechanics to polarity competition. A constant-slope cue also grows in peak with size, helping explain why its length trend differs from the increasing local-patch requirements.	Setting the extension-dependent deactivation coefficient to zero only during stimulation defeats 4/4 native successful gradient pulses. Matching the ramp peak defeats 3/4. Under fixed/relative patches, two longer sampled cells still reverse after the same deactivation-term removal.	Measure physical extension, global activity and local incumbent/challenger histories during the same characterized cue. Compare constant local support with increasing spatial coverage; test a specified mechanical or activity-feedback perturbation while recording its actual effect on shape.

Inference scope

Holmes 3: Fixed-dose controls and matched-state placement support spatially structured competition; mobility is not a universal rescue. Relative support changes exposure. Orientation success, strength restoration and finite local fate are different readouts.

Corrected deBelly: Endpoint budgets explain coupled trajectories; the separate evolved ablation establishes a contribution to resting mechanics. Neither proves a universal MCA withdrawal threshold or a causal share of reversal duration. Low-length decreases and L39 require separate treatment.

Spring: Evolved controls establish geometry-dependent contributions at fixed native doses, not counterfactual duration thresholds. Endpoint motion, geometric dilution and native recovery remain; this is not removal of all mechanics.

Source: mechanism_crosswalk_v2.json · Exact assays, gates and source citations accompany the full legend.